Huawei Access Node JAY-ACC-35 Routing Protocol Flap Resolution SOP

Node: JAY-ACC-35 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_5789

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure on Huawei Access Node JAY-ACC-35 at the Jayanagar Metro Site. The underlying cause of this failure domain is rooted in the dynamic nature of routing protocols, which can lead to an excessive number of routing updates and protocol flaps. This, in turn, can overwhelm the node's control plane, causing packet loss, increased latency, and ultimately, service disruptions. The root cause of the failure domain can be attributed to various factors, including misconfigured routing policies, inadequate routing protocol timers, or hardware-related issues affecting the node's forwarding plane.

Pre-requisite Checks

- Remote hands dispatch to the Jayanagar Metro Site to physically access the Huawei Access Node JAY-ACC-35

- Verify the status of redundant links to ensure no other issues are affecting the network

- Check the current routing protocol configuration and timers to identify potential misconfigurations

Potential Root Causes

Cause	Probability
Misconfigured routing policies causing excessive routing updates	High
Inadequate routing protocol timers leading to protocol flaps	Medium
Hardware-related issues affecting the node's forwarding plane	Low

Diagnostic Commands & Expected Outputs

```
Command: display current-configuration | inc bgp, Expected Output: bgp process-id neighbor [neighbor-address] remote-as [asn], Description: This command will display the current BGP configuration and help identify any misconfigured routing policies or inadequate routing protocol timers.

Command: display bgp process information, Expected Output: BGP process information:, Description: This command will display the BGP process information, including the number of routing updates and flaps.

Command: display ip routing-table, Expected Output: Routing table:, Description: This command will display the routing table and help identify any issues with packet forwarding.
```

Resolution Steps

Step 1: Step: Verify and correct routing policies, Command: undo bgp policy policy-name, Description: Verify the routing policies and correct any misconfigurations to prevent excessive routing updates.

Step 2: Step: Adjust routing protocol timers, Command: bgp process-id timer update-interval [new-interval], Description: Adjust the routing protocol timers to prevent excessive protocol flaps.

Step 3: Step: Check and replace hardware components, Command: display hardware component [component-name], Description: Check the hardware components and replace any faulty components to prevent hardware-related issues.

Post-Check Verification

```
Command: display bgp process information, Expected Output: BGP process information:, Description: Verify that the BGP process information has stabilized and the number of routing updates and flaps has decreased.

Command: display ip routing-table, Expected Output: Routing table:, Description: Verify that the routing table has stabilized and packet forwarding is working correctly.

Command: ping [neighbor-address], Expected Output: Success, Description: Verify that the network is reachable and packet forwarding is working correctly.
```

Roll-back Procedures

- Step: Revert routing policies, Command: bgp policy policy-name, Description: Revert the routing policies to their original state if the fix fails.

- Step: Revert routing protocol timers, Command: undo bgp process-id timer update-interval, Description: Revert the routing protocol timers to their original state if the fix fails.

- Step: Replace hardware components, Command: replace hardware component [component-name], Description: Replace any faulty hardware components if the fix fails.